Magnetic Earth Models – Teacher Copy

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In this class, students use magnetic Earth models to investigate Earth's magnetic field and then create a diagram showing features of the field based on their observations.

Earth & space science curriculum link:

Recognise Earth is a planet in the solar system and identify patterns in the changing position of the sun, moon, planets and stars in the sky (AC9S2U01).

Identify how forces can be exerted by one object on another and investigate the effect of frictional, gravitational and magnetic forces on the motion of objects (AC9S4U03).

Pre-requisite knowledge:

- Students will have encountered magnets before and understand that magnets have two poles (a north and a south) and a magnetic field.
- They may have done basic magnetism experiments, such as placing iron filings around a bar magnet, before.
- Students will understand that the Earth has two geographic poles and that locations are defined by latitude and longitude.

Aim:

By the end of this class, students should be able to:

- Recognise that the earth has a simple, dipolar magnetic field similar to that around a giant bar magnet.
- Make simple observations about how the strength and direction of the magnetic field changes with latitude.
- Draw a simplified diagram of the Earth's magnetic field based on their experimentation and label key features.

Materials (Figure 1):

<u>For the magnetic Earth model:</u>

- Earth stress ball e.g., from Officeworks /Spotlight OR polystyrene sphere e.g., from Spotlight, around 1-3" diameter
- N-S bar magnet e.g., from Aussie Magnets OR a pair of neodymium disk magnets e.g., from Bunnings
- Strong glue (if using disk magnets)

Note: Make sure that your ball is large enough to fully conceal the bar magnet inside.

<u>For the magnetic field detector:</u>

- Pencil with eraser top
- Push pin
- Paper clip
- Flat nosed pliers and/ or wire cutters

<u>Other equipment:</u>

- Closed staples
- Stainless steel scourer e.g., from Woolworths (optional)
- Paper plate/ tray
- Scissors
- Field compass e.g., from Anaconda (optional)
- Pens/ pencils/ coloured pencils

Figure 1: Materials for the Magnetic Earth Models class.

Safety considerations:

- Keep strong magnets away from credit cards, mobile phones, computers and other technology to avoid potential damage.
- Remind students to handle scissors, staples and other sharp craft objects carefully to avoid injury to other students/ staff.
- Consider cutting the holes into the balls and glueing the magnets in place prior to the class so students don't have to handle sharp knives and/ or strong glue.

Activity 1: Constructing the magnetic earth models & field detector

Arrange the class into small groups of around 3 or 4 (depending on how many sets of equipment are available). Each group needs a full set of equipment to make their own magnetic earth model and field detector.

Step 1: How to Assemble a Magnetic Earth

Demonstration video: Magnetic Earth Models Step 1: How to Assemble a Magnetic Earth

If using a bar magnet:

- Use a sharp knife to cut a slit in the ball at one of the poles large enough to insert the magnet inside.
- Push the magnet into the ball until it is fully concealed.

Note: if using an earth stress ball, cut the slit in the southern hemisphere near Antarctica and insert the magnet south-end first so that the geographic north pole is a magnetic south pole.

If using a pair of disk magnets:

- Use a knife or scissors to gouge holes at opposite poles of your ball. The holes should be slightly larger than your magnets.
- Ensure the disk magnets are inserted into the holes with opposite poles on the surface so that they act as a dipole.
- Use super glue or another strong glue to fix the magnets in place.

Note: if using disk magnets, choose a ball with a smaller diameter (around 1").

Step 2: How to Assemble the Magnetic Field Detector

Demonstration video: Magnetic Earth Models Step 2: How to Assemble the Magnetic Field Detector

1. Use pliers to straighten out the paperclip.
2. Bend the paperclip around the pliers so that it has a loop in the centre, as shown in **Figure 2.**
3. Trim the ends of the paperclip using wire cutters so that it is approximately 3 cm in length.
4. Place the end of the push pin through the loop in the paperclip and push this into the eraser at the end of the pencil as shown in **Figure 2.**
5. Make sure the paperclip shape is free to spin between the eraser and pin.

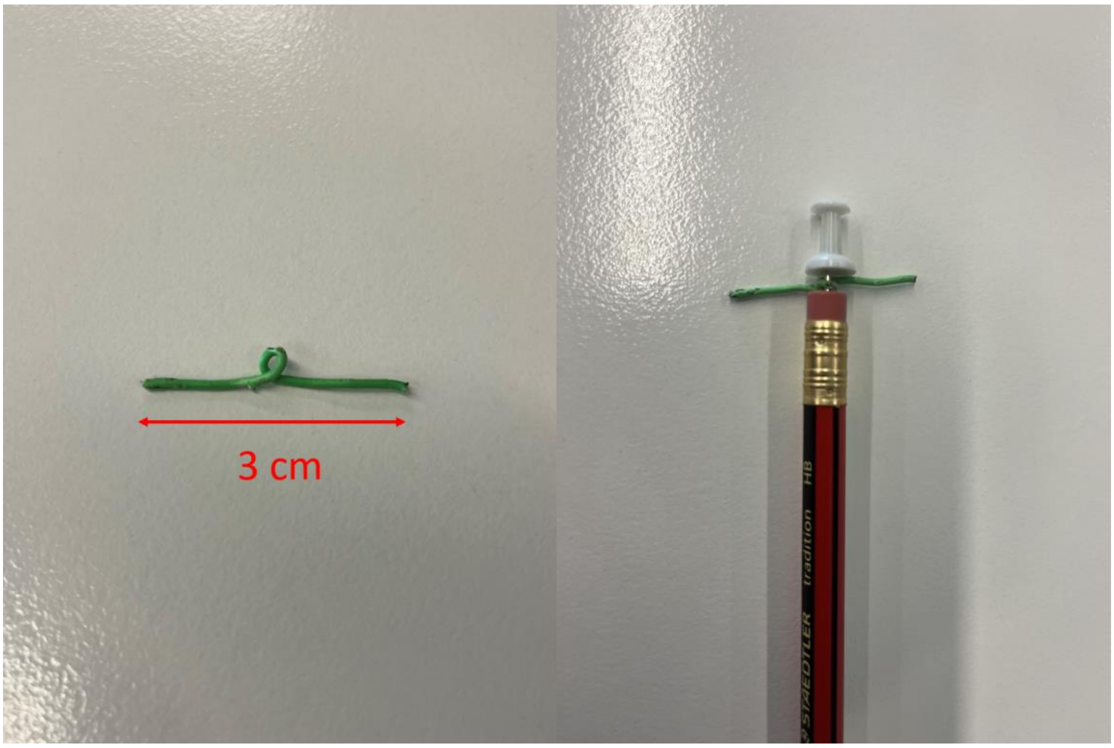

Figure 2: Paper clip bent into shape (left) and completed field detector (right).

Activity 2: Experimenting with the magnetic field around the magnetic Earth balls

Step 3: Using the Magnetic Field Detector

Demonstration video: [Magnetic Earth Models Step 3: Using the Magnetic Field Detector]

- Students should slowly move the field detector around the model Earth from pole to pole.
- They should pay attention to how the orientation of the paperclip changes with latitude, from perpendicular to Earth's surface at the poles **(Figure 3)** to parallel to the surface at the equator **(Figure 4)**.
- They can also experiment with the distance between their detector and the model Earth to investigate how far the magnetic field extends.

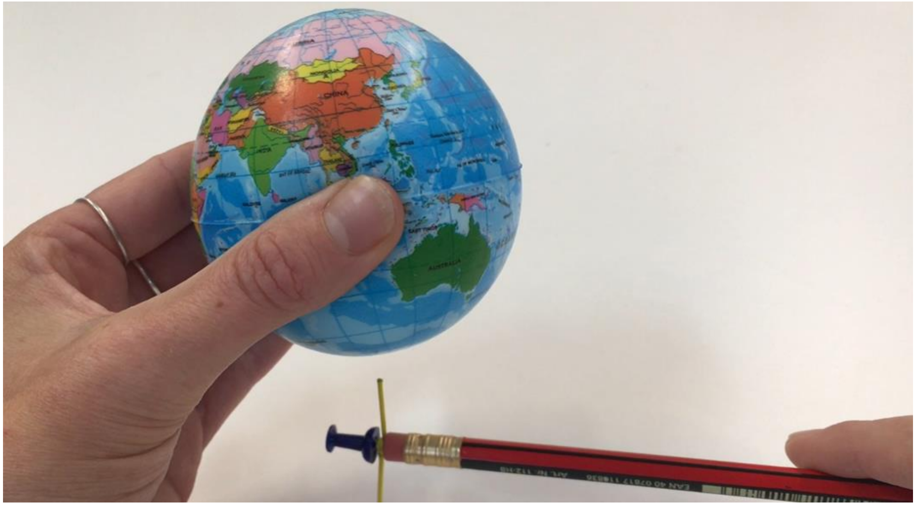

Figure 3: Paperclip is perpendicular to Earth's surface at the poles.

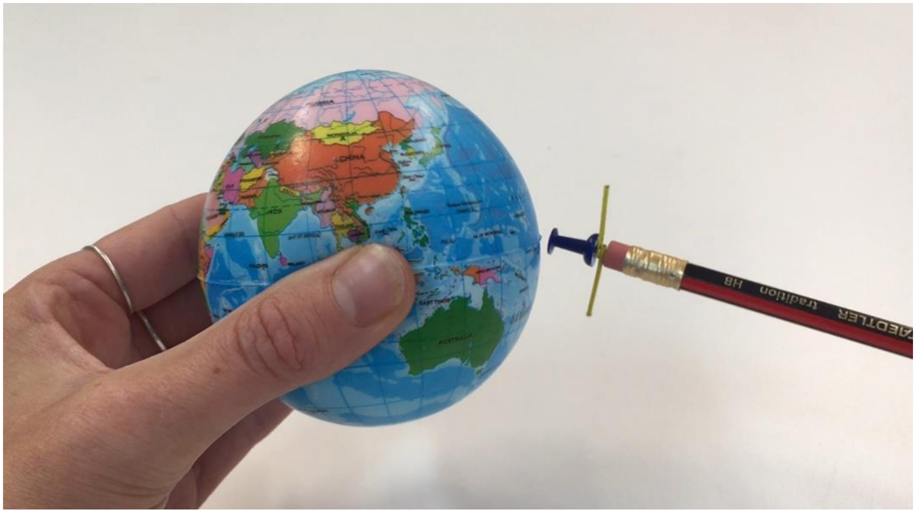

Figure 4: Paperclip is parallel to Earth's surface at the equator.

1. Where does the field detector point at the Earth's poles? How about the equator?

The detector points vertically down/ up into Earth at the poles (perpendicular to Earth's surface) but is parallel to the surface at the equator.

2. What happens when you move the field detector around the Earth?

The detector moves from a vertical orientation at the poles and then flattens towards the equator.

3. Where do you think the magnetic field strength is the strongest – at the poles or the equator?

At the poles (the field detector has a stronger attraction to the Earth at the poles).

Experimenting with a Field Compass (optional)

- If field compasses are available, students can move these around the model, similar to the field detector and observe how the orientation of the compass needle changes.
- The compass needle is perpendicular to Earth's surface at the poles **(Figure 5)** and parallel at the equator **(Figure 6)**.

Note: If using an Earth stress ball, students will notice that the north end of the compass needle is attracted to the northern hemisphere (**Figure 5**). This shows that the geographic north pole is a magnetic south pole (and vice versa for the geographic south pole, which is a magnetic north pole).

Figure 5: A field compass needle is perpendicular to Earth's surface at the poles.

The red north end of the compass needle is attracted to the northern hemisphere – this shows that the geographic north pole is a magnetic south pole.

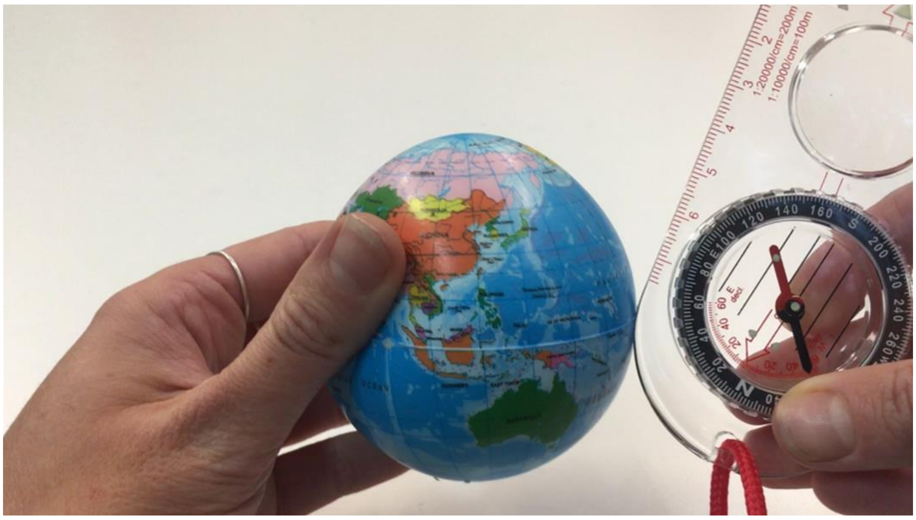

Figure 6: A field compass needle is parallel to Earth's surface at the equator.

Bonus Question: Can you identify the north and south magnetic poles of your Earth?Students should recognise that the north end of a compass needle is attracted to a magnetic south pole (and vice versa for the south end).

If this concept is too complex for students, omit this step and consider removing field compasses from the class.

Step 4: Visualising the Magnetic Field using Staples

Demonstration video: Magnetic Earth Models Step 4: Visualising the Magnetic Field using Staples

- Create a handful of closed staples using a stapler (this can be done by the teacher ahead of the class to save time). Place these on a paper plate/ tray to avoid mess.
- Students should drop/ pour or roll the staples onto the model.
- Students should notice that the staples are perpendicular to Earth's surface at the poles and parallel at the equator. They should also notice that staples concentrate at the poles where the magnetic field is strongest (**Figure 7**).
- Student should try and get their staples to form loops/ chains encircling the Earth from pole to pole (**Figure 7**).

Figure 7: Staples are perpendicular to Earth's surface at the poles, parallel to the surface at the equator, and form chains from pole to pole.

Experimenting with Scourer Pieces (optional)

- As an alternative/ addition to staples, students can model the magnetic field using small pieces cut from a stainless steel scourer (the teacher can cut the pieces ahead of the class to save time).
- Similar to the staples, students should drop/ pour/ roll these onto their model and observe where they stick. The pieces should concentrate at the poles. They should sit perpendicular to the surface at the poles, but parallel to the surface at the equator (**Figure 8**).

Figure 8: Scourer pieces sit perpendicular to the surface at the poles but parallel at the equator.

4. Where do the staples/ scourer pieces want to stick – at the centre of the Earth or the poles?

Staples/ scourer pieces should concentrate at the poles (where the magnetic field is strongest).

5. How does the orientation of the staples/ scourer pieces differ between the poles and the equator?

Staples/ scourer pieces sit perpendicular to Earth's surface/ vertical at the poles but parallel to the surface/ flat at the equator.

6. Can you get your staples to join up and form loops/ chains? Why do they do this?

Students should be able to see the staples joining together as chains from pole to pole and should attempt to connect them as a loop surrounding one side of their Earth (see **Figure 7**).

Staples join up like this because they become magnetised by the main magnet(s) inside the ball.

Activity 3: Drawing the Earth's magnetic field

Provide students with pens/ pencils/ coloured pencils so they can draw onto the template provided in the student worksheet.

Ask students to compare their drawings with those from another group before showing them the following diagram (**Figure 9**).

Features to look for in student's drawings:

- Geographic North and South poles labelled
- Bar magnet in the middle of the Earth labelled with the South magnetic pole in the northern hemisphere and the North magnetic pole in the southern hemisphere
- Field lines (with arrows) enveloping Earth and going from magnetic North to magnetic South. Field lines concentrate at the poles.

The diagram students draw on the template should look like **Figure 9** below.

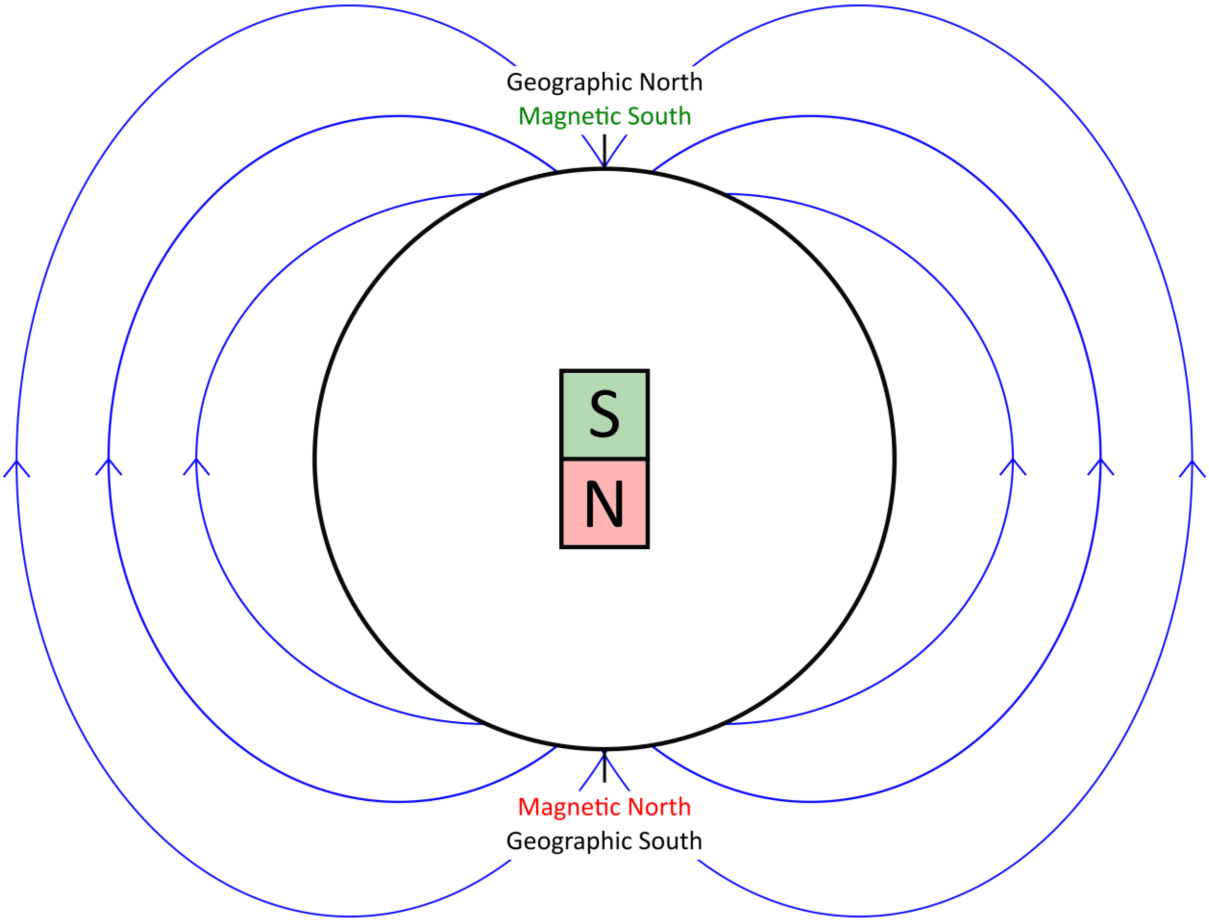

Figure 9: Simple diagram of Earth's magnetic field.

This activity was inspired by:

Arbor Scientific (2009). *Illustrate the Earth's Magnetic Properties | Magnetic Globe | Arbor Scientific*. Available at: https://www.youtube.com/watch?v=iEAl2ZLWB0E&list=PLNVQAZf80b3i1a3PWX2jk8nwNcwQxFq5j&index=4

Lunar and Planetary Institute (2016). *Neato-Magneto Planets*. Available at: https://www.youtube.com/watch?v=vETEE_SPi_M

Marbles Kids Museum (2021). *Magnetic Fields*. Available at: https://www.youtube.com/watch?v=NTbb_9mGYNc&list=PLNVQAZf80b3i1a3PWX2jk8nwNcwQxFq5j&index=6

NASA (2023). *Solar System Magnetism*. Available at: https://www.nasa.gov/stem-content/solar-system-magnetism/

Physics with Simon Poliakoff (2024). *Modelling the magnetic field around the Earth*. Available at: https://www.youtube.com/watch?v=uBnJW9UMZ2o&list=PLNVQAZf80b3i1a3PWX2jk8nwNcwQxFq5j&index=5

Standford University (2015-2020). *A Teacher's Guide to Magnetism and Solar Activity*. Available at: https://solar-center.stanford.edu/activities/Magnetism-Solar-Activity/